- Part 3: Metaheuristics -

Numerical Methods and Optimization

May 7, 2024 — ENSIA

- Combinatorial Optimization Problems -

- This course aims to present the nature and the power of widely used **metaheuristic** methods, primarily those used in **artificial intelligence** and **operations research**. The methods to be covered are used to solve search, reasoning, planning and general engineering **optimization problems**.
- The graduate students who will take this course may use many of the algorithms introduced in this course in their graduate research studies.

Definitions

Objective Function: Max (Min) some function of decision variables

Subject to (s.t.) equality (=) constraints inequality ($\leq$ $\geq$) constraints

Search Space Range of values of decision variables that will be searched during optimization. Often a calculable size in combinatorial optimization

Type of solutions

- A solution to an optimization problem specifies the values of the decision variables, and therefore also the value of the objective function.
- A **feasible** solution satisfies all constraints.
- An **optimal** solution is feasible and provides the best objective function value.
- A **near-optimal** solution is feasible and provides a superior objective function value, but not necessarily the best.

Continuous Vs Combinatorial

- Optimization problems can be continuous (an infinite number of feasible solutions) or combinatorial (a finite number of feasible solutions).
- Continuous problems generally maximize or minimize a function of continuous variables such as min $4x + 5y$ where x and y are **real** numbers
- Combinatorial problems generally maximize or minimize a function of discrete variables such as min $4x + 5y$ where x and y are **countable** items (e.g. integer only).

Definition of combinatorial optimization

- *Combinatorial optimization is the mathematical study of finding an optimal arrangement, grouping, ordering, or selection of discrete objects usually finite in numbers.*

  - Lawler, 1976

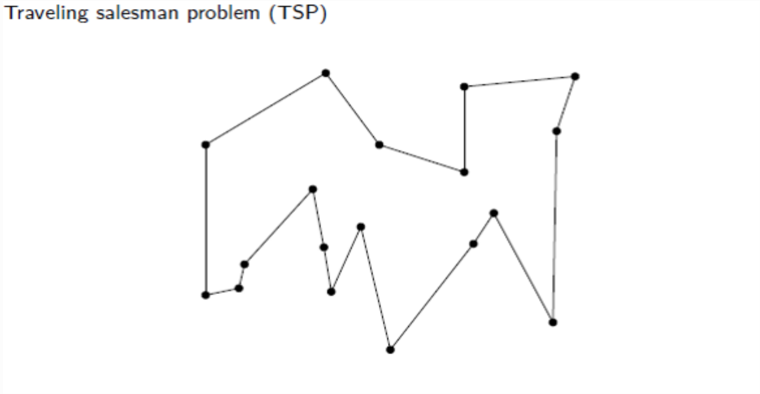
- In practice, combinatorial problems are often more difficult because there is no derivative information and the surfaces are not smooth.

Constraints

- Constraints can be **hard** (must be satisfied) or **soft** (is desirable to satisfy).
- Example: In your course schedule a hard constraint is that no classes **overlap**. A soft constraint is that no class be before 10 AM.
- Constraints can be **explicit** (stated in the problem) or **implicit** (obvious to the problem).

TSP (Traveling Salesman Problem)

Given the coordinates of n cities, find the shortest closed tour which visits each once and only once (i.e. exactly once).
Example: In the TSP, an implicit constraint is that all cities be visited once and only once.



Aspects of an Optimization Problem

- Continuous or Combinatorial
- Size – number of decision variables, range/count of possible values of decision variables, search space size
- Degree of constraints – number of constraints, the difficulty of satisfying constraints, the proportion of feasible solutions to search space
- Single or Multiple objectives

Aspects of an Optimization Problem

- Deterministic (all variables are deterministic) or Stochastic (the objective function and/or some decision variables and/or some constraints are random variables)
- Decomposition – decomposite a problem into series of problems, and then solve them independently
- Relaxation – solve problem beyond relaxation, and then can solve it back easier

Comparaison

Simple problem	Hard problem
Few decision variables	Many decision variables
Differentiable	Discontinuous, combinatorial
Single modal	Multi modal
Objective easy to calculate	Objective difficult to calculate
No or light constraints	Severely constraints
Feasibility easy to determine	Feasibility difficult to determine
Single objective	Multiple objective
Deterministic	Stochastic

- For Simple problems, enumeration or exact methods such as **differentiation** or **mathematical programming** or **branch and bound** will work best.
- For Hard problems, differentiation is not possible and enumeration and other exact methods such as math programming are not computationally practical. For these, **heuristics** and **metaheuristics** are used.